STAT/MA 41600
In-Class Problem Set #26: October 19, 2026
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Problem Set 26 Answers

1a. Because X and Y are not defined in a rectangular region, they are not independent. Alternatively, note that if we multiply the pdfs of X and Y in parts **(1b)** and **(1c)**, their product is not the same as the joint pdf.

1b. The probability density function of X is $f_X(x) = \int_0^{5-5x/8} 1/20\, dy = (1/32)(8-x)$ if $0 < x < 8$, and $f_X(x) = 0$ otherwise.

1c. The probability density function of Y is $f_Y(y) = \int_0^{8-8y/5} 1/20\, dx = (2/25)(5-y)$ if $0 < y < 5$, and $f_Y(y) = 0$ otherwise.

2a. Yes, X and Y are independent, because they are defined in a rectangular region, and their pdfs multiply to yield their joint pdf.

2b. We have $f_X(x) = \int_0^{\infty} 14e^{-2x-7y}\, dy = 2e^{-2x}$ for $x > 0$, and otherwise $f_X(x) = 0$.

2c. We have $f_Y(y) = \int_0^{\infty} 14e^{-2x-7y}\, dx = 7e^{-7y}$ for $y > 0$, and otherwise $f_Y(y) = 0$.

3a. We have $1 = \int_0^5 \int_0^5 k(25-xy)\, dy\, dx = (1875/4)(k)$, and therefore $k = 4/1875$.

3b. No, X and Y are not independent, because their pdfs do not multiply to give their joint pdf.

3c. The pdf of X is $f_X(x) = \int_0^5 (4/1875)(25-xy)\, dy = 4/15 - 2x/75$ for $0 < x < 5$, and otherwise $f_X(x) = 0$.

By the way, by symmetry, the pdf of Y is $f_Y(y) = 4/15 - 2y/75$ for $0 < y < 5$, and otherwise $f_Y(y) = 0$.

4. Yes, X and Y are independent because they are defined in a rectangular region, and their pdfs multiply to give the joint pdf of X and Y: $f_X(x)f_Y(y) = (2/25)(5-x)(1/2)(2-y) = (1/25)(5-x)(2-y)$ for $0 < x < 5$ and $0 < y < 2$, and otherwise $f_{X,Y}(x,y) = 0$.

5. Now consider two random variables X and Y with joint probability density function $f_{X,Y}(x,y) = 63e^{-2x-7y}$ for $y > x > 0$, and otherwise $f_{X,Y}(x,y) = 0$.

5a. No, X and Y are not independent, because we have $Y > X$.

5b. We have $f_Y(y) = \int_0^y 63e^{-2x-7y}\, dx = \frac{63}{2}(e^{-7y} - e^{-9y})$ for $y > 0$, and otherwise $f_Y(y) = 0$.